

Complete API Endpoints Table - 100% Coverage

Category	API Name	Method	Endpoint	Description	Request Parameters	Response Type
SYSTEM ENDPOINTS						
Root	Root Info	GET	/	Welcome message & API info	None	JSON: message, version, ml_enabled, docs
Health	Health Check	GET	/health	Service status & ML availability	None	JSON: status, service, ml_enabled
USER MANAGEMENT						
Users	Create User	POST	/users/	Register new user	Body: username (str), email (str)	UserSchema
Users	Get User	GET	/users/{user_id}	Get specific user details	Path: user_id (str)	UserSchema
Users	List Users	GET	/users/	List all users with pagination	Query: skip (int, default=0), limit (int, default=100)	List[UserSchema]
Users	User Preferences	GET	/users/{user_id}/preferences	Get user behavior analytics & preferences	Path: user_id (str)	JSON: user info, preferences, categories, ratings
ITEM MANAGEMENT						
Items	Create Item	POST	/items/	Add new product to catalog	Body: title, description, category, brand, price, in_stock, product_status, category_name, product_rating, discount	ItemSchema
Items	Get Item	GET	/items/{item_id}	Get specific item details	Path: item_id (str)	ItemSchema

Items	List Items	GET	/items/	List items with filters	Query: skip (int=0), limit (int=100), category (str, optional), in_stock_only (bool=False)	List[ItemSchema]
Items	Similar Items Basic	GET	/items/{item_id}/similar	Get similar items (category-based) [ITEM-BASED REC]	Path: item_id (str), Query: limit (int=20)	JSON: target_item, similar_items, total_found
Items	Similar Items Enhanced	GET	/items/{item_id}/similar/enhanced/	ML-powered similar items [ITEM-BASED REC]	Path: item_id (str), Query: limit (int=20)	SimilarItemsResponse
INTERACTIONS						
Interactions	Record Interaction	POST	/interactions/	Track user-item interaction	Body: user_id (str), item_id (str), interaction_type (str: view/click/purchase/like), rating (float: 1.0-5.0)	JSON: message, id
SEARCH ENDPOINTS						
Search	Enhanced Search	POST	/search/enhanced/	Advanced search with facets & filters	Body: query, category, min_price, max_price, min_rating, in_stock_only, sort_by, limit, offset, filters	SearchResponse
Search	Semantic Search	POST	/search/semantic/	ML semantic search using embeddings	Body: query (str), limit (int=20), plus all SearchRequest fields	SearchResponse
Search	NLP-Enhanced Search	POST	/search/nlp-enhanced/	Natural language search with intent	Body: query (str), limit (int=20), plus all SearchRequest fields	SearchResponse
RECOMMENDATION ENDPOINTS						

Recommendations	Enhanced Recommendations	POST	/recommend/enhanced/	Multi-algorithm recommendations [USER-BASED REC]	Body: user_id (str), algorithm (str: auto/collaborative/content/hybrid), limit (int=20), offset (int=0), context (dict, optional), exclude_items (list, optional)	RecommendResponse
Recommendations	Deep Learning Recommendations	POST	/recommend/deep-learning/	Neural network recommendations [USER-BASED REC]	Body: user_id (str), limit (int=20), plus all RecommendRequest fields	RecommendResponse
ANALYTICS ENDPOINTS						
Analytics	Trending Items	GET	/analytics/trending-items/	AI-predicted trending products	Query: window_days (int=7), limit (int=20)	JSON: trending_items with item, trend_score, interaction_count
Analytics	Popular Items	GET	/popular/	Most popular products	Query: limit (int=20)	List[ItemSchema]
ML MANAGEMENT						
ML Admin	Train Models	POST	/ml/train-models/	Trigger ML model training (needs 100+ interactions)	None (runs as background task)	JSON: message, current_interactions
ML Admin	ML Status	GET	/ml/status/	Check ML system status	None	JSON: ml_enabled, engines dict, models_directory, endpoints_available

Summary Statistics

- Total Endpoints: 20
- GET Endpoints: 10

- POST Endpoints: 10
- User-Based Recommendations: 2 (Enhanced, Deep Learning)
- Item-Based Recommendations: 2 (Similar Basic, Similar Enhanced)

Recommendation Type Classification

Type	Endpoints	Use Case
USER-BASED	/recommend/enhanced/, /recommend/deep-learning/	Personalized recommendations for a specific user
ITEM-BASED	/items/{id}/similar, /items/{id}/similar/enhanced/	Find products similar to a given item

This is the complete, 100% comprehensive list of all APIs in your recommendation engine based on the main.py file you provided.